ABHINAV VERMA

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Education

R.V. College of Engineering, Bangalore, India
- *Electrical and Electronics | CGPA: **7.75***

Guru Nanak Public School
- *CBSE (Class XII) | Aggregate: **87.4%***

B.N.S.D Shiksha Niketan Inter College
- *UP Board (Class X) | Aggregate: **91.6%***

Technical Skills

Programming & Development :
C, C++, Java, Python, C#, JavaScript, SQL, HTML, CSS, ReactJS, Spring Boot

Core Concepts & Technologies :
OOP, Data Structures, Computer Networks (CN), Machine Learning (ML), LLM, RAG

Tools & Platforms :
MATLAB, Simulink, PySpark, Power BI, Visual Studio (.NET), Photoshop, Keil, Proteus, PSpice, Microcontrollers

Work Experience

Stealth | *Application developer (ML)*
- Built compliance-focused agentic AI platform using RAG (OpenAI, Pinecone) for context-aware responses
- Developed secure frontend (React, Shad CN) with user interaction flows
- Designed scalable backend (Node.js, MongoDB, S3) with vector-based retrieval and storage

Tektronix | *Software Engineer (Research and Development)*
- Automated PCIe PLL bandwidth testing using Python & Selenium, replacing costly BERT with AWG-based solution for R&D
- Implemented AWG-based signal generation & filtering integrated with Tektronix RX via VM for test execution

Astilbe Page | *Software Engineer (AI)*
- Built AI newsletter platform (React, Node, MongoDB) with JWT auth & dashboard
- Implemented waitlist/email automation (rate limit, admin, CSV, Node mailer); deployed (Render//Atlas).
- Engineered intent-based LLM routing (Mistral-7B) with fallback logic
- Designed a production-ready dark-theme UI with animated React components, AI chat (session history), and markdown rendering.

Projects

Know BTC: OHLC Analysis || *Python, NumPy, Pandas, Plotly, Matplotlib, scikit-learn, XGBoost*
- Processed and analyzed Bitcoin OHLC data with candlestick visualizations to identify price patterns and trends.
- Performed time-based analysis (monthly, quarterly, yearly) to uncover variations and market behavior insights.
- Applied XGBoost and Random Forest models for price prediction, using feature engineering and evaluating with RMSE, R^2.

Predict-or: A Stock Price Predictor || *Python, NumPy, Pandas, Plotly, Matplotlib, scikit-learn, Tensorflow*
- Preprocessed financial time-series data (cleaning, outlier handling, feature scaling) for model readiness.
- Implemented Random Forest and LSTM models for price prediction, achieving ~98.6% and ~98.9% accuracy respectively.
- Evaluated models using regression metrics and compared performance, with LSTM outperforming RF on sequential trends.

Book Matchmaker || *HTML, CSS, JS, Python, NumPy, Pandas, scikit-learn*
- Implemented popularity-based and collaborative filtering recommendation models using user ratings and book metadata.
- Designed responsive web interface using HTML, CSS, and JavaScript for interactive book recommendations.
- Processed and structured dataset to generate personalized recommendations using similarity metrics.

Campus Wave || *React, Axios, Django, DRF, PostgreSQL*
- Built college management system with React frontend and Django REST API backend.
- Implemented RESTful CRUD operations (students, courses, professors) using Axios.
- Optimized data storage using PostgreSQL for performance & scalability.

Coding Profile

- **Hacker rank** – (Link)
- **GFG** (400+ problems solved) – (Link)

Certificate

- Introduction of Machine learning and its projects –AWS (Click here)
- Joy of Computing Python – **NPTEL Elite Silver Medal** (Click here **)**

Position of Responsibility

- **CARV Hindi**
  Work as a Writer and Actor and Perform various street plays and theater plays across Bangalore and India.
- **Teaching assistant at UPAY NGO**
  Handled Math and English for the students at NGO.

Achievements & Extra Curriculum

- Qualify National Talent Search Exam (NTSE) Stage I host by Government of India (GOI).
- Winner in **Street Play competition** at Jain University, Bits Pilani Goa (Oasis).